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Second Life Patient Scenarios: Enhancing the Diversity of the Nursing Profession

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Despite the current transformation of the U.S. population to one of increased diversity, the majority of nurses employed are from White backgrounds. In addition, few secondary school students from ethnically diverse backgrounds or from rural regions in the nation are encouraged to pursue a collegiate education or to consider nursing as a career. This article describes an innovative 3-year program in which one rural university in the southeast used a virtual environment, Second Life, to expose secondary students to nursing through role-playing as avatars interacting with patient case scenarios. Secondary school science and health occupations teachers developed patient and environmental scenarios under the direction of the nursing program director and the information technology director from the university. Throughout the 3 years, 300 rural and multicultural students were exposed to virtual world health care learning through quizzes with five patient case and environmental scenarios. Student and teacher evaluations were positive about their interest in learning obtained through the patient case examinations. Technical difficulties during Year 1 were eliminated by Year 3. This program demonstrates that use of virtual technologies such as Second Life may increase the interest of secondary rural and multicultural students in careers in nursing and in pursuing a collegiate education.

Keywords: Second Life; diversity; patient scenarios; avatars



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The landmark report of the Sullivan Commission on Diversity in the Healthcare Workforce (2004) stated that the nation's population is becoming increasingly ethnically diverse. The report stated that the nation's nursing workforce must adjust to provide culturally competent care. This recommendation found a "direct link between poorer health outcomes for minorities and the shortage of minority health care providers" (Sullivan Commission on Diversity in the Healthcare Workforce, 2004, p. 2). The Institute of Medicine (2010) has stated that the nursing profession must increase its ethnic diversity to meet the changes in the nation's population. The Sullivan Commission on Diversity in the Healthcare Workforce noted a gap between the minority population of the United States (25%) and the minority population of nurses (9%; p. 2).

Despite the evident need to increase the diversity of the nursing workforce, promoting diversity of students in the nation's nursing programs remains challenging

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(Duerksen, 2013). Culturally diverse students have been found to have lower test scores on standardized examinations for entrance into nursing programs as compared to White students (Gilchrist & Rector, 2007). Olson (2012) found that students for whom English is not their first language had a 40% lower pass rate on the National Council Licensure Exam as compared to White students. Minority and rural students additionally have many issues with maintaining cultural ties with their families, including having to provide funds for their families requiring them to work while in school. Other barriers to success include perceived discrimination, lack of English proficiency, cultural tensions, and difficulty paying for tuition, room, and board (Amaro, Abriam-Yago, & Yoder, 2006).

THE NURSING NETWORK-CAREERS AND TECHNOLOGY PROGRAM: INTRODUCING INNOVATIVE PATIENT CASE SCENARIOS IN SECOND LIFE

Second Life is a highly interactive multiuser virtual environment where millions of people throughout the world have been meeting, communicating, and collaborating for over a decade.

The Nursing Network-Careers and Technology (NN-CAT) Program was established in 2013 at a university in the southeastern United States by faculty who were concerned about the shortage of rural and ethnic minority students being accepted into the nursing school. The program was sponsored by a generous grant from the Health Resources Services Administration. The university is located in an impoverished rural mountainous area with a composition of families with unmet needs, a multitude of issues impacting social and structural determinants of health, education disparities, and socioeconomic barriers. The mountainous region has four distressed counties: Family incomes are 67% less than the national average and poverty rates are 150% or greater than the national average. An ongoing disparity in these communities is the lack of access to transportation to medical appointments; grocery, hardware, and clothing stores; and pharmacies. The population of the NN-CAT Program's region has become more ethnically diverse, with 85% of the population surveyed as White, 1.8% surveyed as African American, 9.4% surveyed as Native American or Alaskan native, 0.9% surveyed as Asian, and 2.7% indicating other (Light, 2014).

The Second Life Health Sciences Program was a component of the main NN-CAT Program. The purpose of this program was to provide a virtual environment and curriculum through the development of patient case scenarios. The program was developed by secondary school health occupations education teachers, an IT specialist from the university, and the Second Life Health Sciences Program nursing program director. The purpose was to administer the case scenarios to students in the five high schools in the counties and the Native American reservation surrounding the university. The Second Life curriculum was upgraded annually at three teacher workshops, each held at the university over 2 days of training in which faculty could use avatars to enter the case scenarios and "care" for the patients. High school faculty consisted of health occupations teachers and physical science teachers who were led through the intricate development of the various patient case and environmental care scenarios set in a rural town. The scenarios developed in the 3 years of the program were

- A middle-aged Native American man with diabetes
- A 16-year-old high school athlete with a respiratory illness
- A 65-year-old woman with cardiovascular problems

Barriers to success for minority and rural students include having to work to provide funds for their families, perceived discrimination, lack of English proficiency, cultural tensions, and difficulty paying for tuition, room, and board.

- A 12-year-old female student with a persistent urinary tract infection
- A 14- year-old high school basketball player who developed skeletal injuries from repeated movement

The environmental case scenarios developed by the science faculty were

- An explosion of a nuclear power plant in the factory located in the mountains near the rural towns in which the students lived
- The noxious effects of smoke from a nearby paper mill

THE EFFECT OF SECOND LIFE ON NURSING EDUCATION

Second Life is a highly interactive multiuser virtual environment where millions of people throughout the world have been meeting, communicating, and collaborating for over a decade (Benham-Hutchins, Lall, & Cohn, 2015). According to Linden Labs, the creator of Second Life, more than 35 million people have registered as members since its launch in 2003, and approximately 1 million people actively use it every month (Linden Labs, 2013). Online users create a graphical representation of themselves called an *avatar* and enter multimedia environments which are intended to replicate real-world settings and experiences (Lee & Park, 2011). Users may customize the appearance of their avatars and interact with virtual worlds that may be beaches, mountains, multiplayer games, concerts, and/or educational programs (Irwin & Coutts, 2015).

The use of Second Life to enhance simulation education has been prominent for the past decade with undergraduate, graduate, and postgraduate nursing students across the globe (Tiffany & Hoglund, 2014). Second Life provides a platform for virtual clinics, health care settings, and environmental scenes to be developed that allow students (avatars) to interact within the virtual world experience. Trangenstein, Weiner, Gordon, and McNew (2010) completed a literature search reviewing 29 nursing education virtual environments and discovered that nursing educators used Second Life for exploring health care issues and for communication through distance learning and simulation activities. Second Life has been integrated into international health care education by the Geneva Foundation for Medical Education and Research (Schoonheim, Heyden, & Wiecha, 2014).

The use of Second Life as a virtual simulation environment for nursing students has been generally well received; evaluations of its use in educational settings have been positive (Irwin & Coutts, 2015). Difficulties with the installation and development of virtual health care settings require the expertise of an information technologist, and computer log-in issues can be complex for students in the early stages of using Second Life for learning (Chow, Herold, Choo, & Chan, 2012).

EVALUATING USE OF SECOND LIFE SCENARIOS WITH MULTICULTURAL SECONDARY STUDENTS

Throughout the 3 years of the Second Life Health Sciences Program, 300 students participated in active learning about the role of the nurse in caring for patients in a simulated hospital. In addition, students cared for patients involved in environmental care scenarios in a rural mountain environment. Demographic data about the students who participated is listed in Table 1.

TABLE 1. Second Life Health Sciences Program Student Diversity Composition

Year 1	Student Composition	Year 2	Student Composition	Year 3	Student Composition
African American	5	African American	3	African American	0
Native American	8	Native American	9	Native American	38
White	53	White	67	White	99
Multiracial	2	Multiracial	5	Multiracial	6
Asian	0	Asian	3	Asian	2
Total	68	Total	87	Total	145
				3-Year Combined: 300 Students	

Throughout the 3 years of the Second Life Health Sciences Program, the total number of students increased by 52.4%. This was attributed to the increase in the number of classes and students that were enrolled at each of the participating high schools. This increase in the number of classes and students was responsible for the rise of White students by 45.4% over the 3-year period. The most significant increase in students was in the Native American classes where enrollment increased 78.9% during the program. The Native American reservation high school had additional classes of students who were enrolled into the program during Year 3. A decline of African American students during Year 3 was attributed to graduation of previous year's students.

Faculty and students were asked to complete an evaluation with 20 questions that were rated as positive or negative about the use of Second Life as a learning tool in their high school. Overall ratings by students and faculty were positive. Students enjoyed the simulated world and preferred it over traditional classroom learning. The students had a rating of 92% positive responses to the 20-question evaluation about the use of Second Life. Faculty rated the experience as being meaningful for the students and one that was stimulating for active learning. The faculty had a score of 93% positive responses to the 20-question evaluation survey regarding the use of Second Life as a simulation learning tool in the high school classroom.

The purpose of the Second Life Health Sciences Program was to evaluate both the students' and the faculty's ability to learn and use a simulated virtual environment with the goal of caring for patients and environmental case studies. This goal was met, according to the evaluations from the students and faculty which indicated that simulation was a viable medium for learning in the high school environment. Additional data about the number of students who pursued college after graduation, such as enrolling in the local community college or university, or graduation rates from each of the participating high schools, were beyond the scope of this project and were not obtained. These data would be valuable to have to further evaluate the benefits of the Second Life Health Sciences Program. Future programs of this type would be encouraged to track graduation and college entrance rates for further evaluation of success.

Lessons learned from this 3-year project include that students from rural and multicultural backgrounds enjoy learning about nursing through a simulated environment such as Second Life. Students were extremely engaged and readily sought the answers to the questions posed by the patients in the scenarios.

The environmental case scenarios developed by the high school science faculty were an explosion at a factory located near the rural towns in which the students lived, and the noxious effects of smoke from a nearby paper mill.

This platform provided an innovative medium for both students and faculty to explore the role of a nurse in today's health care environment in a fun and enjoyable format. Rural and multicultural students were offered a rare opportunity to learn about nursing and encouraged to seek further education to become a health care professional.

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